

Simpson's Paradox Shuffle Student Handout

TOK Cognitive Science Activity Bank • Mathematics • 35-45 min

Category	Mathematics	Time	35-45 min
Difficulty	Medium	ID	simpsons-paradox-shuffle

Big idea: Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

Student-Facing Prompt

Inspect Simpson's Paradox Shuffle Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Simpson's Paradox Shuffle Stimulus A	Student-facing stimulus 1: Method A beats Method B overall, but Method B performs better inside each difficulty subgroup. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Simpson's Paradox Shuffle Stimulus B	Student-facing stimulus 2: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Use this for comparison, a second round, or a partner challenge.
Simpson's Paradox Shuffle Reveal / Twist	Student-facing stimulus 3: Headline card: 'Method A has the higher success rate' beside subgroup table cards. Work the example, then decide whether it gives a pattern, a model, a calculation, or a proof.	Teacher reveal: connect the changed response to the big idea — Aggregated data can hide subgroup structure; mathematical knowledge depends on how cases are grouped.

Actual Lesson Questions

#	Question for students
1	After inspecting Simpson's Paradox Shuffle Stimulus A, what is your first knowledge claim?
2	Which exact detail from Simpson's Paradox Shuffle Stimulus A did you use as evidence?
3	What did you infer beyond the evidence?
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?
5	What missing information would most improve the knowledge claim?
6	How do grouping choices shape statistical knowledge?

Ready-to-Use Examples

- Method A beats Method B overall, but Method B performs better inside each difficulty subgroup.
- Headline card: 'Method A has the higher success rate' beside subgroup table cards.

Student Task

Use the stimulus cards to complete the observation/inference/evidence/confidence grid, then answer one TOK knowledge question connected to Mathematics.

Activity Steps

- 1. Give groups cards for two methods split across two subgroups.
- 2. Students calculate or compare success rates inside each subgroup.
- 3. Groups combine the cards and calculate the overall rate.
- 4. Reveal the reversal and ask which claim is more responsible.
- 5. Debrief why a mathematically accurate headline can still mislead.

Observation and Evidence Record

Use this grid to keep observation, inference, evidence, and confidence separate.

What I noticed	What I inferred	Evidence	Confidence

TOK Debrief

- Knowledge question: Can correct calculations produce a misleading knowledge claim?
- When should data be aggregated, and when should it be separated?
- How do categories and grouping decisions shape mathematical evidence?

Knowledge Questions

- How do grouping choices shape statistical knowledge?
- What makes a data comparison fair?
- Can a correct average hide a more important truth?

Transfer Example

For example, run this activity with a simple classroom version using subgroup success-rate cards for two fictional study methods.. Have students commit to an initial claim, then reveal the change, hidden rule, alternative context, or comparison that makes the knowledge problem visible.

Exit Ticket

- One thing I first treated as evidence was...
- My confidence changed because...
- A better knowledge question I can now ask is...